GLACIER MELT: EMBODIED KNOWLEDGE AND THE COMPOSITION OF CHOREOGRAPHIC SOUNDSCAPES

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ABSTRACT

This paper presents an exploration on the role of embodied knowledge in composition process of soundscapes for choreography, utilising both subjective reflection and analytical techniques. Using Glacier Melt, a soundscape by Mark Pedersen for a choreographic work-in-progress by Prue Lang, the authors examine their own processes for aligning sound and movement, and compare these reflections to the subjective responses of dancers and audience members, as well as objective measures of the sound and movement material of the work in order to propose ways of cultivating greater awareness of and access to embodied knowledge in the composition process.

1. INTRODUCTION

The making of new artistic work always has a period of development, which frequently includes specific experiments and research into new techniques and processes. It was in the context of such a research and development phase that the authors collaborated, culminating in a limited public showing of a work-in-progress by Prue Lang that is part of her larger *Five Senses* project. The research questions Lang is posing within this development phase is to what degree can choreography be used to deliberately explore or provoke cross-sensory experiences, and what is the impact of such cross-sensory experiences on the reception of the choreographic work?

From a sound design perspective, the research questions of interest centre on the development of techniques for designing an aural environment for choreographic performance which are congruent to the affective goals of the choreography, and more specifically in developing a deeper understanding of the process of intermodal art making.

Frequently in sound design for choreography, there is a stark contrast between the processes used for art making. Dance making is, by nature, physical and much of the communication between the choreographer and the dancers is firstly via observation, demonstration and adjustment of physical moves, positions, and postures, secondarily underpinned by mental images and processes which are verbally communicated. In contrast, the sound design process is typically mediated through a computer interface and related interface tools, either in a studio environment or the rehearsal space, and as such the sound designer is frequently stationary, and at a distance from the dancers and choreographer. Communication is more frequently verbal or written, and furthermore, constrained by the lack of common terminology. At times, rather than sound designer and choreographer being able to work coherently toward a common goal, the sound design process is akin to stumbling in the dark hoping to find the light switch.

Embodied knowledge can be understood as being the type of knowledge in which the body knows how to act (Tanaka 2011), and is distinct from instinct or reflex, in that it typically arises from physical repetition (practice) and may sometime be referred to as muscle memory. Given the central role of embodied knowledge in the creation of choreographic work, the question then arises as to whether the tools and techniques of sound design can accommodate the transmission and use of embodied knowledge to overcome some of the communicative barriers in working across the different modes of expression within the two art forms. This paper combines subjective reflection by the authors with dancer and audience responses along with material analysis of the performance of the work-in-progress in order to explore this question.

2. CHOREOGRAPHY AND SOUND DESIGN

In the broader framework of contemporary choreography, sound design has taken on a significantly different role to that of the traditional musical score in ballet. Most famously, John Cage and Merce Cunnigham deliberately disconnected choreography and composition in their collaborative work. Likewise in the choreography of William Forsythe, ``music and sound are no longer relegated to the background of a dance but instead run independently in parallel to the organizing of bodies'' (Slater 2011).

In the work under discussion, the objective of the sound design is to support the choreography without distracting from it or competing with it. Which is not to say that the sound is always congruent to movement. Lang has a clear process of developing the choreography first, in isolation from the final sound design, and then introducing the sound design late in the development process. Frequently what is sought is sound design which

offers a counterpoint to the movement, while still providing points of connection.

3. THE MAKING OF ….

3.1. Choreographic Context

Glacier Melt was composed as a soundscape for Part 1 of Five Senses, a five minute improvisation for four dancers, which featured slowly evolving responses to contact, at both an intra- and interactive level (i.e. contact between one body part and other and contact between dancers). Conceptually, one body part/dancer initiates contact and the other part/dancer receives information from the contact, but responses follow unexpected pathways.

The overall style of movement in this piece could be characterised as sustained, free flowing and indirect, with a tendency toward a feeling of heaviness. This mood is punctuated by moments of quick, light, direct movement that swiftly return to slow flow.

The choreography was well developed before the final sound design commenced, and in fact had not been seen by the sound designer before the composition of Glacier Melt.

3.2. Sound Design Process

The starting point for the design of Glacier Melt was a textual description of the choreography:

> "A soft very slow & fluid part with the hand sliding duets"

along with some suggestions for the sound quality:

> "something soft and meditative, circular … something that enhances/compliments"

Although the descriptive content of the text is quite sparse, this was sufficient to bring to mind a very clear sonic image for the piece. The sound design was completed in approximately 30 minutes based on two takes of improvisation with a combination of audio software (Native Instruments Reaktor) and modular outboard effects (Mutable Instruments Clouds). The resulting piece consists primarily of slowly cascading sine tones with occasional instances of harmonic distortion and delay.

Reflecting on the composition process, the role of embodied knowledge was significant. The brief textual description of "hand sliding duets" was familiar from the authors' previous collaborations, and evoked a physical sensation, a mental image of the movement, and with those, a clear sense of what the piece should sound like.

While prior collaborative experience helped with the interpretation of the text, personal experience of somatic practices such as Tai Chi, yoga and Shintaido were much more dominant in shaping the internal awareness of the energetic and affective qualities that were sought in the sound design.

In this sense, the very sparseness of the textual description was helpful, in that it allowed room for Pedersen's non-textual, embodied knowledge to inform the sound design process.

However, in examining the role of embodied knowledge in the sound design process, it is not particularly helpful to simply rely upon the happy alignment of personal experience and personal intuition of the sound designer with the intentions of the choreographer. The question we pose is whether embodied knowledge can be cultivated in support of sound design for choreography?

4. MOVEMENT, MATERIALISM AND MEANING

While the appeal to embodied knowledge, at first glance, appears to be a retreat into the purely subjective, there are also collective responses to sound that appear common and have their grounding in embodied cognition. The body is a starting point, since as embodied beings, human sound cognition is intimately linked with the way our brains also process movement, and as such, embodied sound cognition suggests that there are common elements to sound that underlie cultural and individual subjective layers of interpretation For in depth discussion of embodied music cognition, see Leman (2008) and Godøy (2010).

Three concepts are particularly useful to consider at this point:

- that sound is autonomous of audition, i.e. has its own materiality, independent of subjective experience
- that affect is likewise autonomous of subjective experience, allowing for individual, subjective experience while not denying the existence of a non-subjective reality
- that sound carries inherent meaning from the relationship it has to our own physical movement

This aspect of the relationship between sound, movement and embodied knowledge is recognised by Julian Henriques in *Sonic Bodies* (Henriques 2011), a detailed analysis of the systems of knowledge that underpins Jamaican sound system practice. Henriques proposes a model of knowledge which places the sensing, kinetic body at the centre of a sonic logos founded on ratio and relationship rather than representation.

Henriques's concept of a sonic logos builds upon Lefebvre's (2004) *rhythmanalysis*, embracing a broader framework of vibration as not just the means of propagating knowledge and affect, but taking the position that affect is vibration. Affect is carried across varying media, be they electromagnetic, corporeal, or

sociocultural, and each of these layers of media manifests the rhythms, amplitudes and timbres of vibrational affect in different ways, be that certain patterns of audible frequencies, pulse rates, muscle contractions, patterns of movement, emotional responses, or cycles of social events, styles or fashions (Henriques 2010).

The same vibrational analysis is also found in Alfred North Whitehead's process philosophy, which underpins much recent significant theorising of sonic experience, from the work of Steve Goodman (Goodman 2009) to the work of Susanne Langer, as discussed by Priest (2013).

The vibrational-materialist approach places the locus of meaning within the relational web of vibrating and listening bodies. Affect finds its meaning in the relational, multi-sensory matrix of repeated patterns, arising from the repetition of a musical phrase, the enactment of ritual, the cycle of seasonal events.

Movement, sound and affect can be considered as different modes of the same vibrational relation - or rather the same vibration moving through different media.

This idea informed the composition of Glacier Melt in a practical sense, utilising prior investigation into qualities of motion and qualities of sound, particularly looking at Laban correlations of effort qualities and Smalley's spectromorphology (Pedersen and Alsop 2012).

Using these concepts as an investigative framework, we sought to look deeper into both the subjective responses and the material properties of the work in order to identify points of contact and transmission – conduits for the flow of affect.

4.1. Subjective Responses

Lang previewed the work at home while viewing a recording of a rehearsal. Glacier Melt was the closest of three offered selections, but she was unsure if it fitted the piece. Only when trialled with dancers in costume, lighting and music at the right level was there a sense of confirmation that the sound and movement matched the quality she sought for the work. The specific criterion for Lang in confirming the selection of Glacier Melt was that the sound design helped in delivering the right *live energy* within the space of the performance – the synergy between the choreography, the dancers, the sound and the space was key.

The dancers responded positively to the soundscape during final rehearsals and performance, with one commenting that the soundscape was ``complementary to the state [of being] the choreographer was looking to achieve… the idea and the music worked to create an atmosphere; personally this allowed me to find a better mind body connection: something to enter into as opposed to something [I had] to create''.

Audience responses to sound design for choreography are often sparse, and in as much as the intention for this piece was that sound should support but not dominate the choreography, this is to be expected and even welcomed. In the context of the performance under discussion, which was presented in a salon style setting in which the audience comments on the works at the end of the performance, audience responses were collected via informal verbal comments during the feedback session, and were followed by written email communication upon request.

Audience responses included noting that the performance and the sound both "felt cold" and "gave the impression of being under water". One member of the audience specifically noted the impact of the sound design, noting that they were struck by what they perceived to be "a perfect confluence of sound and movement … the constantly sliding pitch, neither opening nor settling in any particular tonality, and the way the movement flowed and slid and oozed in (apparent) response gave the impression that we, as an audience, were deposited directly into the middle of something that had been occurring in our absence...and would continue when we were gone again."

4.2. Objective Analysis

Figure 1 gives an overview of the sound and motion in terms of labelled sonic features overlaid on a spectrogram, with the amplitude wave form is shown below and a motiongram of the dancers' performance is shown above.

The motiongram (Jensenius 2006) was derived from a video recording of the performance, and shows the average percentage of motion within each 1 pixel wide horizontal band of the original video. Darker areas of density represent greater amounts of movement. Within the overall view in Figure 1, you can see changes in the amount of motion at different heights over time, with greater density of motion being at around torso height for most of the performance, and some obvious periods of greater motion low down (closer to the floor) just prior to the 2 minute mark and again just after the three minute mark.

Figures 2 & 3 compare more detailed excerpts from both the spectrogram and motiongram analyses. While there are no direct correlations between sound events and movement events, the qualities of each excerpt show some similarities, with the earlier material in Figure 2 being energetically calmer than that in Figure 3. Note the somewhat more intense harmonic strands and the presence of frequently upwardly rising spectral sweeps in the spectrograph component of Figure 3, indicating a significantly more complex soundscape at this point than in Figure 2. Likewise note the greater density of movement in the motiongram, including more significant vertical shifts in Figure 3 compared to Figure 2.

By way of contrast, Figure 4 shows an excerpt of a motiongram from Part 2 of Five Sense, featuring a more typical style of contemporary dance solo with many abrupt transitions and articulations. The motiongram reveals the much more dynamic style of performance in Part 2 compared to Part 1, with sudden gestures producing spikes and swoops in the motiongram. There was no musical accompaniment for this part of the choreography.

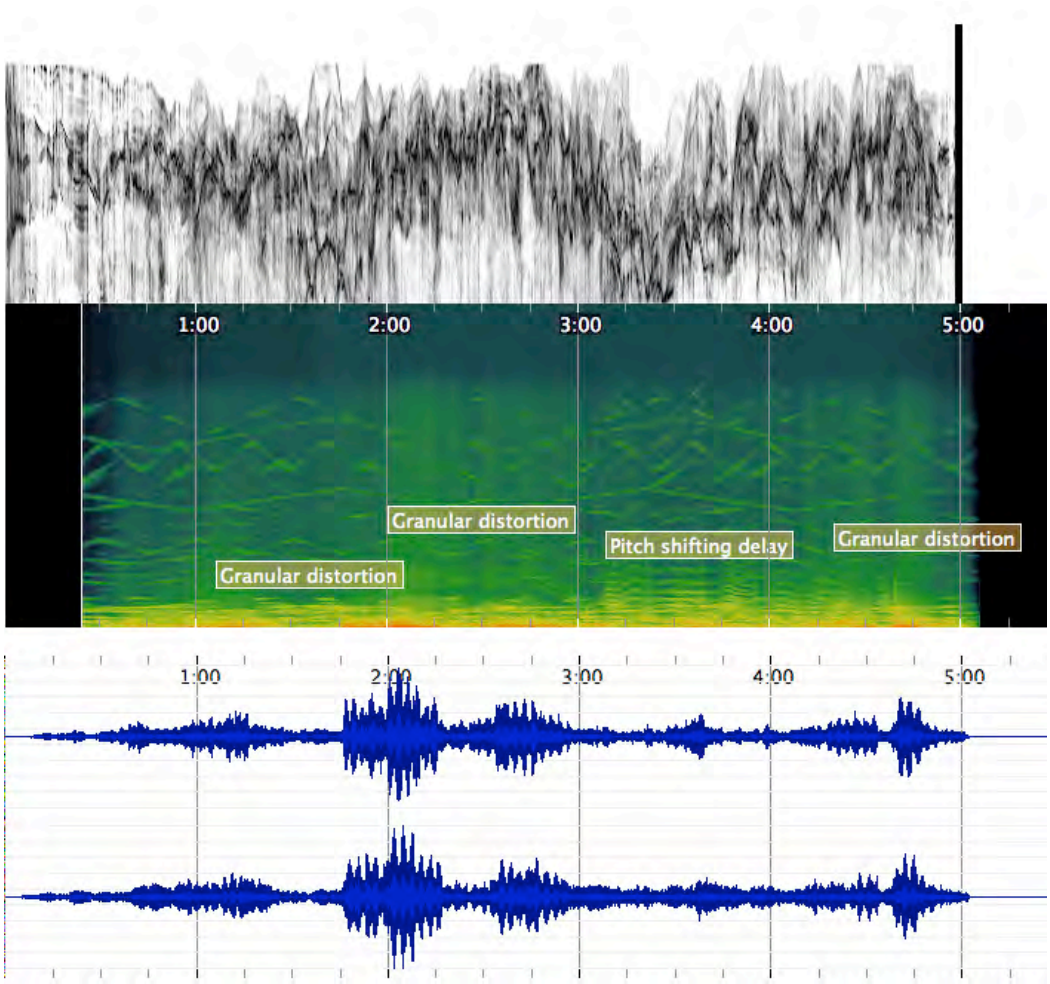


Figure 1. Spectrogram analysis of Glacier Melt aligned with motiongram of Five Senses (Part 1).

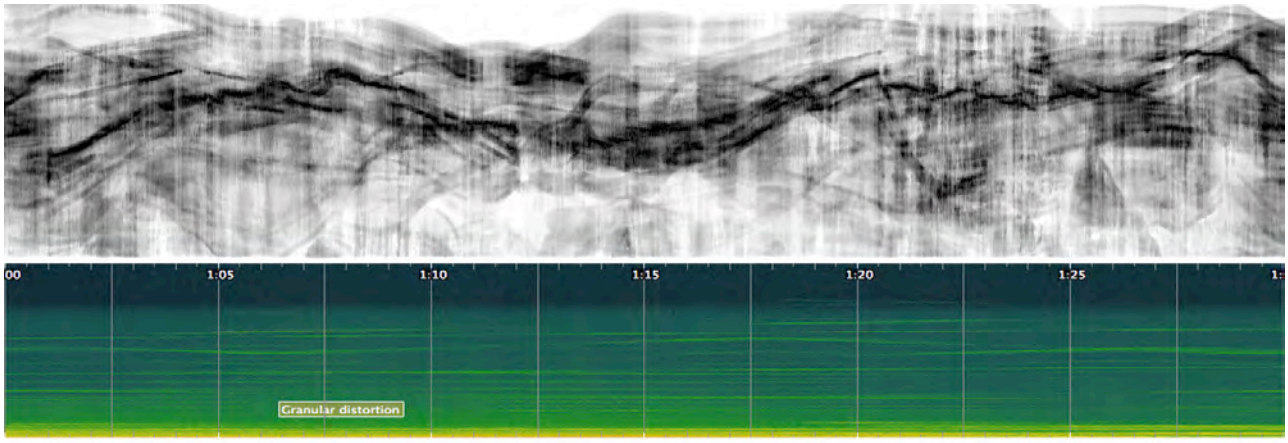


Figure 2. Spectrogram-motiongram excerpt 1 (1:00 – 1:30).

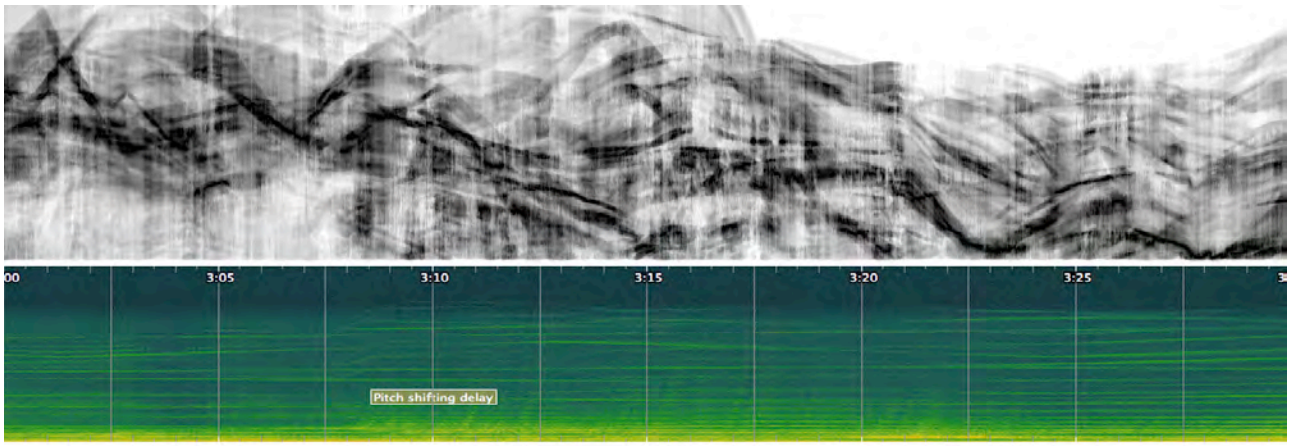


Figure 3. Spectrogram-motiongram excerpt 2 (3:00 – 3:30)

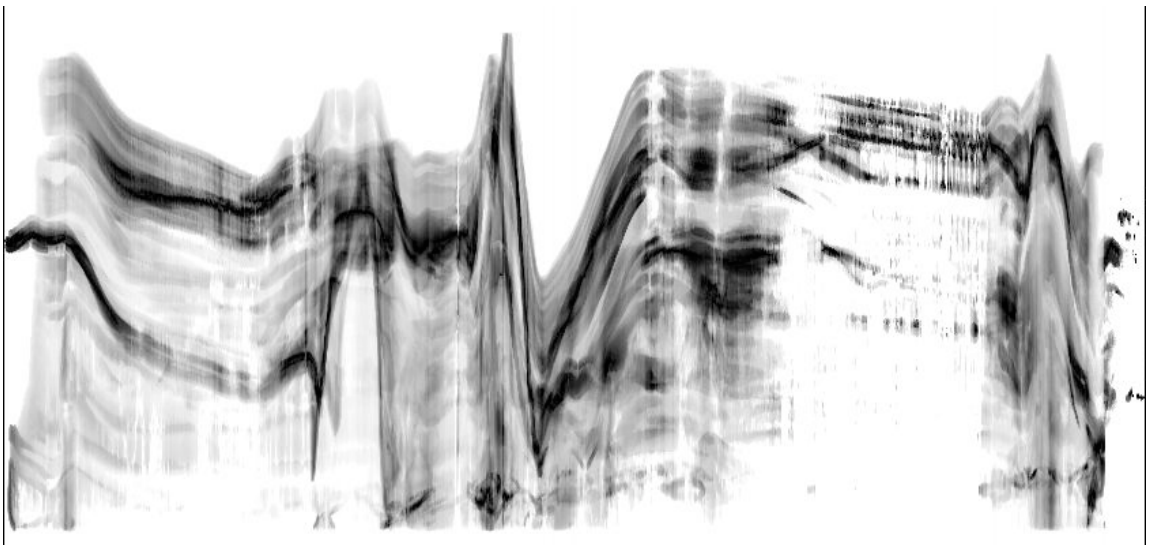


Figure 4. Motiongram excerpt from Part 2 (0:40 – 1:20)

4.3. Extrapolating from the abstract

Given the challenges in communicating verbally across modalities, the degree to which artistic collaborators share physical experience of each other's art forms helps to overcome the paucity of verbal communication. However, relying upon the embodied knowledge of dance that a sound designer may bring with them is inherently limiting. As noted in Section 3.2, there was a happy alignment of somatic experience and the choreographic content in Part 1 of Five Senses that enabled the composition of Glacier Melt to be informed by a congruent internal physical sense of the affective intent. For Pedersen, this degree of embodied knowledge would have been lacking if the sound design had been required for Part 2.

It is here that objective observations and analysis of one modality can help inform the creation process in another. By abstracting the complex physicality of the dancers' movement through motiongram analysis, qualities of motion can be observed and reflected upon in isolation from the choreography itself. This can be particularly useful if there is a preference for not constructing the sound design as a direct response to the choreography, as may occur if composing with direct reference to a video recording of a performance.

The motiongram in Figure 4 provides an insight into the dynamic qualities of that specific choreography, which can inform the sound design in ways that otherwise might not be possible, on account of either a lack of common verbal or notational language for communicating the style of movement or the desired sound qualities, or a lack of somatic experience with that style of movement.

While this approach to cross-modal knowledge sharing can be taken literally, and indeed, some research has been conducted into directly sonifying motiongrams, taking the motiongram as kind of generated score for the sound design (Jensenius 2013), we propose that an alternative approach is to *re-embody* the knowledge communicated abstractly by the motiongram.

The challenge with any re-interpretation of abstract knowledge lies in the mapping from abstraction back into a concrete form. In sound design, this can take the form of mapping (abstract) gestures to specific synthesis parameters, or, as Jensenius does, mapping from one graphic representation into a spectrogram and resynthesizing the generated spectrogram directly. In both cases, there can be much difficulty in the selection of appropriate synthesis parameters and the design of appropriate translations functions.

Working on the assumption that human beings are excellent transducers of affect, and that this is in large measure a function of embodiment, the next step in our exploration of cross-modal art making is to experiment with the use of motiongrams as guide for gaining personal somatic experience that is congruent with choreography as part of the sound design process. The goal is to enable the creation of strongly congruent pieces like Glacier Melt for choreographic material which might otherwise be difficult for the sound designer to assimilate.

5. CONCLUSIONS

We consider this just the beginning of examining the question as to whether the tools and techniques of sound design can accommodate the transmission and use of embodied knowledge to overcome some of the communicative barriers in working across the different modes of expression within sound design and choreography. By examining the subjective congruence of Glacier Melt as a soundscape for a specific choreographic form, we've found through the comparison of spectrograms and motiongrams what appears to be a reasonable objective analysis which is supportive of, but does not necessarily explain this congruence. The goal is not to follow a reductive process of seeking directly translatable metrics from one artform to another, but rather to find useful ways of understanding the subtle qualities of choreography and sound design in order to support the creation of cross-modal artworks.

To this end, further investigation is needed to test how these objectives measures can be used to support sound design for unfamiliar modes of movement (i.e. where embodied knowledge may be not be present to support the sound designer's understanding of the movement or the affective intent). One approach to this would be to use motiongrams to extract gesture profiles which could be explored by the sound designer in the context of their own practice.

This line of investigation will focus on whether the general observation of similarity at the energetic level in the objective data is sufficient to support the goal of creating other cross-modal works which achieve the same degree of congruence found in Glacier Melt. In particular, we propose to investigate ways in which such information is used not in direct translation from one material to another, but rather used to cultivate embodied knowledge within the practitioners, using the body as an integral part of the creation process in both choreography and sound design.

6. REFERENCES

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